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SURVEY

Routing Protocols and Mobility Predictions for UAV Data Collection From Agricultural Sites: A Survey

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ABSTRACT Unmanned Aerial Vehicles (UAVs), also known as drones, have found applications in precision agriculture owing to their agility and remote access. However, because of battery, bandwidth and environmental constraints, UAV's networks experience intermittent links and packet drops. These constraints can either force the UAV to abort its mission or increase packet drops. Therefore, there is a need to improve routing protocols and UAV path planning. This work reviewed routing protocols such as reactive and hybrid as well as mobility predictions such as regressive and Markov models with the central argument of topology control and path optimizations. Optimizations minimize the coverage time and energy consumption. As routing protocols operate on the network layer, this review analyses articles that include cross-layer issues, modelling, and predictions. This article also examined QoS metrics (such as bandwidth and link quality), antenna selection and placement on the UAV and the methods used to validate routing protocols. In this work, UAV's path optimizations consider crop locations, the appropriate placement of ground sensor nodes and how neighbouring UAVs are controlled for efficient data collection and relay. Analysis of the collected data can improve crop health, cultivation management, livestock monitoring, and reduction on operational cost. Therefore, this work resolved that by predicting the topology change, link-lifetime, and residual energy, the routing protocol can reduce retransmission and packet drops, thereby enhancing network longevity.

INDEX TERMS Changing topology, data collection, precision agriculture, mobility predictions, routing protocols, unmanned aerial vehicles.

I. INTRODUCTION

Unmanned aerial vehicles (UAVs), are agile and exhibit a 3D motion that translates into desirable features such as the ability to hover and fly over inaccurate and hazardous terrains while avoiding obstacles. These features make deployment easy, enabling UAVs to be used in many commercial applications, including precision agriculture [1].

Despite their desirable characteristics and increasing popularity, UAVs have inherent limitations that impede their potential application [2]. These constraints include limited coverage range owing battery constraints, intermittent wireless links owing to mobility, and limited bandwidth. Second, and specific to agriculture, there are environmental challenges such as weather (wind gust, extreme temperatures and precipitation), the terrain of the area that calls the UAV to manoeuvre into valleys and across hills [3], and seasonal variation in vegetation cover. Wind causes turbulence and instability, leading to inappropriate coupling of radio signals

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with the antenna [4], especially when the antenna is not rigid and poorly oriented with respect to the antenna radiation pattern as well as its polarization. Vegetation attenuates communications between UAVs or between UAV and ground sensor nodes [5]. The dynamic nature of the network results in routes frequently becoming invalid [6], in addition to increased signalling and data overheads. Frequent network partition and data retransmissions translate into energy holes, bandwidth wastage, high latency and network inefficiencies [7] because the routing protocol exchanges routing information frequently. The combined effect of terrain, mobility, weather and vegetation variation results in large and small-scale fading [8], [9], and image blurring [10]. The fading and link intermittency result in network faults, partitions, and packet drops [11].

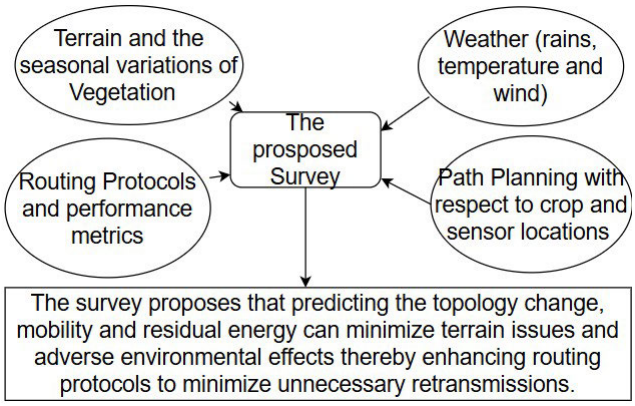

FIGURE 1. Conceptual framework.

As UAVs gain popularity in precision agriculture and the challenges mentioned above hinder efficient UAV data collection, this article reviewed routing protocols and mobility predictions with the central argument of improving network reliability and efficiency. The data collected from the farm could be in the form of pictures and videos captured directly by the UAV. Alternatively, pictures, videos and other data such as relative humility, soil fertility and soil texture are captured and stored in repositories [12]. The collected data can be analysed for decision-making, such as crop monitoring, livestock monitoring, and cultivation management [13]. This in turn leads to a reduction in farm maintenance costs.

Motivation The inherent limitations of UAV and environmental constraints require that the routing protocol be deployed to minimize unnecessary retransmissions. Proper selection and placement of antennas on UAVs, predicting UAV movements and link lifetime will minimize network partitioning and faults enabling data collection before the UAV is out of range.

The use of drones eliminates the use of long range data transmission relays via multiple hops, which are often costly owing to the deployment of relay hops [14]. In addition, the performance of multi-hop communication degrades with node failure. The cost of operation and task completion is reduced as batteries power the UAVs, and one person can operate it [15].

Scope Routing protocols developed for Mobile ad hoc networks (MANETs) are unsuitable for accommodating UAV networks owing to their computational limits and dynamic nature. Therefore, this study analysed the routing protocols with respect to design metrics such as node mobility, link quality, energy and bandwidth, including the effect of these metrics on routing protocols. Second, manoeuvring complex terrains requires prediction and path planning algorithms with respect to crop location and sensor node points, which is one of the contributions of this work. Seasonal variation in vegetation and extreme temperatures require the proper placement of ground sensor nodes, and the design of algorithms. Thirdly, the effect of wind and precipitation require proper antenna selection and placement on the UAV to avoid inappropriate signal coupling. Therefore, there is a need for mobility prediction, link-lifetime predictions, and careful selection and modification of protocols to minimize energy consumption and avoid the frequent exchange of signalling messages.

The remainder of this paper is organized as follows. The methodology for article selection and exclusion is presented in Section II and Section III provides the background. Section IV presents related reviews, while Section V analyses the routing protocols. Section VI examines UAV applications in precision agriculture and Section VII discusses challenges and future work. Section VIII concludes the survey by presenting the major findings and insights gained from the survey such as improving routing and prolonging network lifetime using predictions.

II. METHODOLOGY

As routing protocols and mobility predictions are affected by link quality, topology changes, energy, and bandwidth, this work referenced and examined articles that demonstrate how these factors affect mobility predictions and routing protocols. The following questions were considered while the reference journals were searched.

1. What messages are exchanged by the protocol for efficient data delivery?
2. How can energy and bandwidth be effectively optimized so that the protocols can operate and complete the task in the required time?
3. Are these protocols designed for specific environments/terrains?
4. What factors cause link intermittencies and how do these factors affect routing protocols and UAV path planning?
5. What gaps must be addressed for the deployment of UAVs in the agricultural domain?

A. CRITERIA FOR INCLUSION AND EXCLUSION

Other criteria for selection are network types, use of artificial intelligence (AI), cross-layer issues, and the effect of modelling and predictions of air-to-ground communications

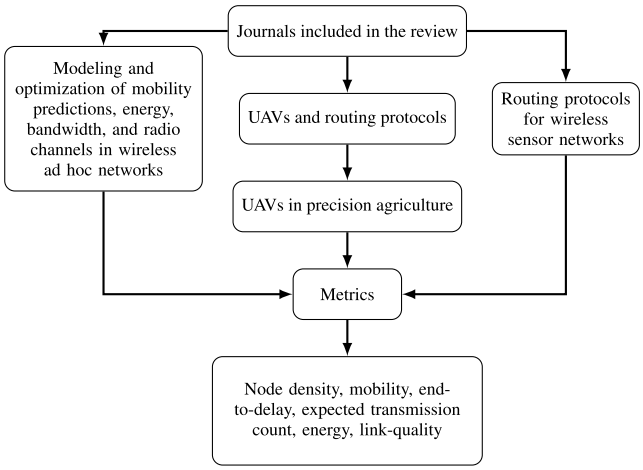

FIGURE 2. Modes of journal inclusion.

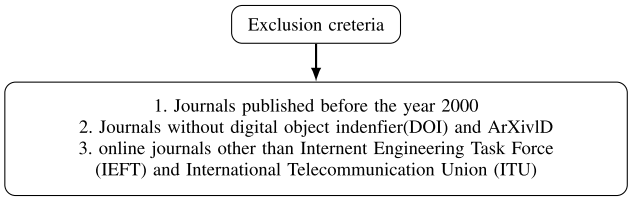

FIGURE 3. Modes of journal exclusion.

TABLE 1. Summery and number of articles used.

Thematic Area	Year 2020 and Below	Year 2021 and Above
Agriculture	04	15
Link Quality	04	16
Mobility and Path Planning	07	13
Energy modelling and Optimization	04	04
Bandwidth	01	01
Routing protocols	24	18
Sub - Totals	43	67
Total	110	

on routing protocols. These data effectively inform the review process regarding future research directions and lead to substantial conclusions.

The selection criteria are not biased in choosing the publishing house as long as the chosen article is relevant to the central arguments of this article and meets the selection criteria as shown in Figure 2 as well as exclusion modes as shown in Figure 3.

B. NUMBER OF JOURNALS STUDIED AND THE YEARS OF THEIR PUBLICATIONS

The reviewed articles are categorized into thematic areas such as Agriculture, Node mobility, Link Quality, Energy and Bandwidth as in Table 1. Most papers that fall under the year 2020 and below are works published on routing protocols for UAVs as well as other ad hoc networks. This clearly justifies that only a few studies have been conducted on the enhancement of routing protocols for UAV applications. Hence there is a need to conduct further research on these protocols. The literature also shows an increasing research in the usage of link quality in network designs as well as the applications of UAVs in Agriculture.

III. BACKGROUND

A. UAV'S NETWORK CHARACTERISTICS

For data collection from agricultural land, a UAV network can be designed as:

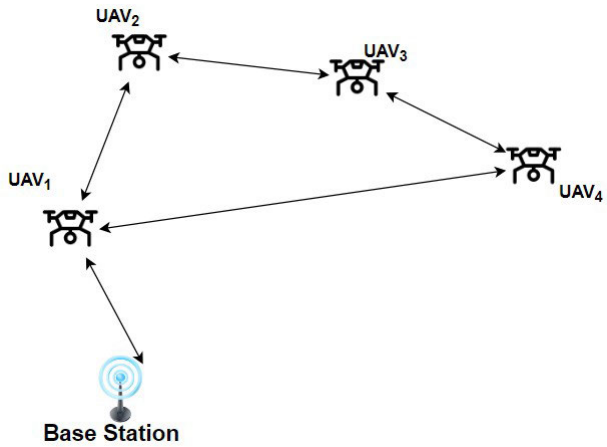

FIGURE 4. UAV swarm connected to base station.

1) SWARM-BASED NETWORK

This is a system of drones that communicate with each other to achieve a common objective as shown in Figure 4. Drones communicate with themselves in addition to a central computer and make decisions to adjust their paths [16] as they observe their environment and neighbour drones. This approach addresses the network disruptions caused by mobility by enabling drones to stay together, distribute tasks, and coordinate activities. The collected data can be relayed to the ground station or central drone. This network is costly to deploy because more drones require more funds.

2) UAV-GROUND SENSOR NODES

In this type of network, a UAV can be used to collect data such as soil moisture and temperature from sensor nodes placed under ground [17] or to collect data of interest from sensor nodes placed above the ground [18]. However, this type of network is limited by environmental constraints such as signal obstruction by vegetation and buildings in rural and urban settings. Routing protocols will be forced to re-establish the link frequently if the modelling of radio resources and path planning is poorly performed.

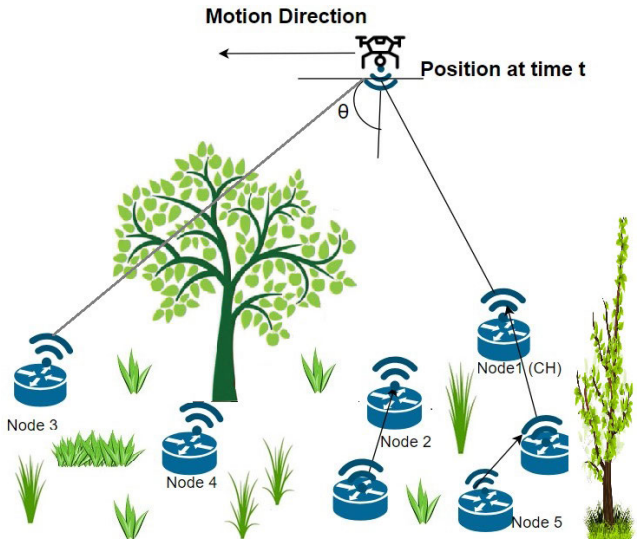

FIGURE 5. UAV - ground nodes network.

B. DESIGN METRICS

For efficient protocol operation and communications between UAV and UAVs or UAV to ground nodes, this study is interested in how the following relates to predictions and routing protocols:

1) NODE MOBILITY

UAV mobility is usually unpredictable, with speeds ranging from 30 km/h to 460 km/h, which results in frequent link breakages and increased signalling overhead as the routing protocols have to retransmit route requests and route errors. These messages consume energy [19] and bandwidth; as such, the algorithm designed for mobility must check for the residual energy and link quality to determine the proximity distance to the target in order to collect the data. Mobility also degrades QoS owing to intermittent [20] links. This calls for design considerations such as predictions, path planning based on ground node locations or crops, and optimizations.

2) LINK QUALITY

This can cause link failure owing to mobility, noise, and interference from other sources. Environmental terrain, including vegetation, can also induce packet loss, frequent disconnections, and network fragmentations. This forces protocols to execute frequent route repairs and re-establishments, drain energy and consume bandwidth [21].

3) ENERGY

UAV are usually on batteries, which results in a reduced network lifetime. The network lifetime of ground sensor nodes depends on their battery energy levels, hence protocols that optimize energy consumption are required. Otherwise, the ground sensor nodes can experience frequent packet retransmission owing to low energy, rendering the protocol unsuitable. As the battery constrains the UAV flight time, this requires optimization and path planning [22] with respect to ground sensor node placements to minimize energy consumption. This article suggests a requirement for energy models that capture parameters such as the initial energy, transmit, receive, idle, and sleep power that the nodes learn to make decisions.

4) BANDWIDTH

This has been used to improve the QoS of many networks. The entire bandwidth of a channel is not available for data transmission because some bandwidths are required for device connections and control overheads. For high-density networks, routing protocols and communication channels suffer from congestion [23] and bandwidth constraints, resulting in network instability and increased packet losses and delays.

IV. RELATED REVIEWS

This survey analysed related work and extracted their major contributions and open issues as summarized in Tables 2 and 3 below. These two characteristics inform the work on gaps that have not yet been addressed.

V. ROUTING PROTOCOL

Topology-based Protocols use source-to-destination information stored in the routing table, which shows the availability and characteristics of the links along the route. These are IP-based protocols and are subject to network interruptions [31]. The effect of node mobility, link quality, energy and bandwidth on these protocols are summarized in Table 4, which includes the following categories:

A. STATIC PROTOCOLS

Here, the computation of the routing tables is performed at the start of the tasks, and the tables are not updated during the operations. These protocols are applied to static or slow-changing networks, and the average source-to-destination links do not change. They are not fault-tolerant, hence, they are unsuitable for applications in a changing environment [32] because they cannot accommodate changes in topology. Examples include data-centric routing (DCR), multi-level hierarchical routing (MLHR), and load carry and deliver (LCAD).

1) DATA-CENTRIC ROUTING (DCR)

Data aggregation occurs before transmission in sensor networks. DCR protocols [33] usually query certain areas of the network such that the sensors or UAVs located in those areas reply with messages that are used to autonomously control the topology. Message producers decide which type of data must be generated and disseminated [34] and the data can be used for path planning by considering factors such as rapid response and signal fading.

Effect of Bandwidth, Energy, Link Quality, and Mobility This protocol works well in cluster topologies such as the swarm-based network shown in Figure 4, but faces energy

TABLE 2. Summary of existing related reviews in UAV networks.

Article	Major attribute and quality	Open Issues
[24]	Presented protocols for wireless cooperation and collaborations among UAVs in an ad hoc manner. The article further analysed design considerations such as topology, mobility, latency, prediction, flight formation, collision avoidance, external disturbances and prediction to achieve scalability, adaptability and robustness	Frequent link breakages, hybrid performance metrics, evaluation tools such as NS2/3, OPNET, OMNET++, security issues
[25]	Presented a comprehensive survey of the architecture, constraints, mobility models, routing techniques and the tools for the simulations. They further detailed and performed a comparative studies on classification of the existing protocols used for FANETs	Areas that have not been exhaustively addressed include: Recharge stations to address energy issues, radio propagation models, and security attacks
[26]	A justifiable AI - enabled routing algorithms that accommodate highly dynamic network topology were reviewed. The AI integrates computational intelligence into different aspects of the UAV networks by assisting the UAV perceive networks and the overall conditions of the environment. The AI also enables fast processing of information in real time	Limited payload, processing power and the dynamic nature of the network calls for design of edge - cutting machine learning algorithms.
[27]	A comprehensive review of UAV networks and the communication standards for UAV to UAV and UAV to ground links. Routing protocols reviewed are the proactive, reactive, hybrid and static. The work discussed that these protocols can be enhanced by adding extensions or modifying control messages to adopt them to the scenario at hand. Mobility models were also reviewed to tailor the UAV for specific applications hence flexibility and better performance.	The dynamic nature of the network results into link disconnections. Others are energy issues and simulation tools selections.
[28]	The article reviewed the characteristics and types of the links in FANETs with comparisons to other ad hoc networks such as MANETs and VANETs. It also identified high adaptable and scalable with high residual energy, bandwidth and low latency as design requirements for the routing protocols	The high number of exchanged messages results into high energy consumptions. High mobility causes frequent link disconnections.
[29]	Thoroughly described the different strategies and routing techniques and then classifications of the routing protocols for agriculture applications such as crop surveying and mapping, crop scouting and insurance as well as cultivation management. It further performed theoretical analysis on routing protocols with the objective of identifying one that can serve better in the agricultural domain subject to mobility models	Choosing the right protocol for the design of multi-UAV systems
[30]	The article presented a system model that includes the network model and mathematical model for UAV - assisted data collection network. It then analysed key technologies that will aid data collection such as 1. clustering of sensor nodes and data collection while hovering or flying the UAV 2. Optimization of path planning and resource allocations such as bandwidth	Multiple access techniques to solve transmission conflicts between competing nodes. Edge - cutting sensors to enable the UAV position itself and minimize errors while collecting data

and bandwidth constraints as part of the bandwidth used to establish connections between the UAVs that are battery constrained. In the case of topology changes, link quality also degrades.

2) HIERARCHICAL ROUTING (MLHR)

This class of protocols is designed to scale energy, number of nodes, and topology changes [35]. They can create a one-level cluster hierarchy across the network or a multi-level hierarchy that can respond to the changing topology. The application of the three types of hierarchical routing examined by [36] in UAV-assisted networks are linear sensor routing, cluster-based routing and tree-based routing. It was found that linear sensor routing reduces the transmission range and number of hops resulting in improved energy usage. Cluster-based routing was found to perform well in a wide area with high node density, whereas tree-based routing minimized the UAV's trajectory distance.

Effect of bandwidth, energy, link quality, and mobility This zonal routing protocol requires optimizing the node coverage and efficiently routing data to the control centres

TABLE 3. Summery of the approaches and design considerations of the related reviews.

Article	Network type	Routing Protocol	Protocol design and modification	QoS met-rics	Validation of protocols	Specific application area	Path planning and optimizations
[24]	UAV swarm	✓	✓	✓	✓	X	Not considered
[25]	Swarm and UAV - ground nodes	✓	✓	✓	✓	X	Considered
[26]	UAV swarm	✓	✓	✓	✓	X	Considered
[27]	Swarm and UAV - ground nodes	✓	✓	✓	✓	Agriculture	Considered
[28]	UAV - ground nodes	X	X	✓	X	IoT	Considered
[29]	Swarm and UAV- ground nodes	✓	✓	✓	✓	Agriculture	Considered
[30]	UAV-ground nodes	Not speci-fied	Not specified	X	✓	IoT	Considered

or repositories; otherwise, with poor optimization, nodes will be detached from the network. Inefficiency in handling the routing protocol results in frequent retransmission and wastage of bandwidth.

3) LOAD CARY AND DELIVER (LCAD)

This protocol relays data between a ground node and a UAV, and then between another UAV or ground node. The LCAD protocol increases the security and data throughput. However, using a single UAV increases the delays when the distance between the source and destination increases. To solve this delay issue, a UAV swarm network is required, which again increases the cost.

Effect of bandwidth, energy, link quality, and mobility UAV mobility directly affects link quality [37], which in turn affects transmission, bandwidth consumption, and energy.

B. PROACTIVE PROTOCOLS

Nodes use tables to maintain the routing information of their neighbours in a given network section. Even if a route is not used, it is preserved. The tables are updated periodically [38]; hence, they are subject to bandwidth constraint, but equally important as the tables contain the latest statistics about the path, there is no need to wait as a path from source to destination can easily be selected. Examples of proactive protocols are:

1) OPTIMIZED LINK-STATE ROUTING

This protocol operates on multi-point relays (MPR) which can cause delay and packet drops when the relay bandwidth is less than the originator bandwidth. Using HELLO messages, a node receives information about a set of nodes that have selected it as their MPR node. The set of nodes has a maximum of 2-hops hence this protocol has and is suitable for UAV data collection. The table is updated once a 2-hop neighbour or MPR node is lost. Other variants of the OSLR are D-OLSR, CE-OLSR, and M-OLSR.

Effect of bandwidth, energy, link quality, and mobility This protocol is suitable for UAV data collection, and it supports up to a maximum of 2-hops; however, owing to the dynamic nature of the UAV network, frequent link breakages exists. Hence, the protocol has to update the routing table frequently [39], which may deplete energy and waste bandwidth.

2) DESTINATION SEQUENCE DISTANCE VECTOR (DSDV)

This is a table-driven routing protocol. The table is consistently updated by broadcasting or multi-casting a routing update packet throughout the network regarding topology changes [40]. The updates contain the latest entries of all possible routes and destinations, and a sequence number is used to differentiate expired routes from the fresh ones.

Effect of bandwidth, energy, link quality, and mobility These updates require energy and bandwidth, which is detrimental to nodes with low energy and causes packet drops and delays. This protocol consumes more energy than AODV owing to frequent topology changes [41].

3) BABEL

Is a distance vector routing protocol that limits the routing loops and the network back holes. Nodes with low energy levels use large time constants for hello messages and update intervals whereas the opposite occurs for nodes with high energy levels [42]. The loop-free feature minimizes the bandwidth usage.

Effect of bandwidth, energy, link quality, and mobility For dynamic topology, the frequent table updates drain the battery and consume bandwidth. Intermittent links compromise the QoS.

4) BATMAN

It does not calculate the graph for the end-to-end path but rather divides the knowledge about the path to all participating nodes to find the best next hop to the destination [43].

The originator node sends an originator (OGM) message to its neighbours, and the neighbours broadcast the message to inform their neighbours of the originator. Each node perceives and maintains only the next-best hop to the other nodes. Hence, knowledge of global topology is not necessary.

Effect of bandwidth, energy, link quality, and mobility Poor quality links, as well as links with low bandwidth, suffer from path loss or delay along the way through the graph. In addition, an intermediary node with insufficient energy causes delays or packet drops, affecting the protocol.

C. REACTIVE PROTOCOLS

The on-demand protocols that generate and update routes between two nodes based on packet availability for transmission. Path finding is repetitive and takes time to establish; hence, high latency may appear before data transfer occurs [44]. The advantage of these protocols is that, they minimize control packets throughout the entire network making them bandwidth-efficient in comparison to proactive protocols. These protocols include:

1) AODV

This protocol obtains routes on-demand with less or no reliance on recurrent advertisements [45]. Unlike dynamic source routing (DSR) where a large overhead is incurred by carrying source routes in the overhead, AODV dynamically establishes route entries at intermediate nodes and uses HELLO messages to detect and monitor links, minimizing overhead.

Effect of bandwidth, energy, link quality and mobility The need for route discovery is initiated by an address resolution protocol (ARP) packet, which incurs unnecessary overheads [46] and consumes bandwidth. Owing to the dynamic nature of the network, the links between the nodes frequently become invalid, resulting in the requirements for route repairs and retransmission, which in turn consumes energy and bandwidth.

2) DYNAMIC SOURCE ROUTING

This is a multi-hop wireless mesh network that is self-organizing and self-configuring. To send a packet to the destination, the protocol uses route discovery, and route maintenance is used to maintain the routes [47]. To minimize the overhead incurred during the route discovery procedure, the protocol uses a cache [48] to save the route path from the sending node to the destination node.

Effect of bandwidth, energy, link quality and mobility Similar to AODV, the dynamic nature of the network results in unstable links, which may in turn result in link failures. These failures may result in retransmissions that consume bandwidth and energy.

D. HYBRID ROUTING PROTOCOLS

Uses reactive protocols during route establishment and proactive protocols when topology changes or when re-establishing a failed link [49]. The following are examples of these protocols.

1) ZONE ROUTING PROTOCOL

This protocol limits the scope of proactive procedure to the neighbours of a node using an intra-zone routing protocol (IARP) to minimize the waste associated with the proactive concept. The global search through the network is performed using inter-zone routing protocol (IERP) loop-free nodes [50] instead of querying all nodes in the network, thereby increasing the reliability and performance. The routing zone of each node is part of the network in which all nodes within a given number of hops can be reached.

Effect of bandwidth, energy, link quality and mobility As in proactive protocols, maintaining the network graph consumes a large amount bandwidth as the tables are updated. The size of the zone is characterized by the number of hops required to reach the border nodes. The larger the distance the larger the number of broadcast messages and the energy required to transmit data and control messages. For better link quality and longer connectivity, the relative speed between the sender and receiver nodes must be minimized [51]; otherwise the network will partition.

2) AODV-BR PROTOCOL

To maintain connectivity in a UAV network, a module based on Boids of Reynolds was added to AODV [52] to control UAV movements. This module enables the UAV to avoid disconnections by adjusting its speed to meet the average speed, and avoiding being too close to one another, and avoiding obstacles. This protocol minimizes the route discovery process by maintaining active paths as mobility is maintained for a while.

Effect of bandwidth, energy, link quality and mobility In the case of a link failure or a sudden departure of a node from the swarm [53], reconstructing the graph requires a large number of control messages as the working mechanism of AODV. This consumes bandwidth and energy.

3) TORA

A link reversal algorithm that is adaptive, efficient, scalable and loop-free protocol. This protocol limits message propagation in a highly dynamic environment [54] by adapting to local topological changes and reducing of communication overhead. It exhibits multipath routing capability, which minimizes congestion in case of node failure [55]. This multipath capability also improves bandwidth utilization and ensures reliable packet delivery, thereby reducing the energy consumption incurred through retransmissions.

Effect of bandwidth, energy, link quality and mobility As the use of a multipath strategy can reduce energy consumption, latency and bandwidth wastage, mobility increases the link breakages which in turn increases control messages [56]. As the node density increases, the routing

traffic sent by the TORA increases [57]. The increase in control messages and routing information consumes bandwidth and energy and increases end-to-end delay.

4) LINK-STATE ESTIMATION-BASED PRE-EMPTIVE ROUTING PROTOCOL (LEPR)

This protocol employs the importance of GPS location information in addition to link quality, safety degree and mobility prediction to account for the past, present and future states of the link [58]. It is based on the AODV routing protocol and uses the link quality stability metric to estimate the link disjoint path and path discovery phase. Through pre-emption, the protocol finds and switches to reliable routes early before failure occurs, thereby reducing packet latency.

Effect of bandwidth, energy, link quality and mobility

The link quality, safety degree, mobility prediction and pre-emption approaches reduce overheads which in turn improves bandwidth utilization, however they do not consider the effect of congestion in the path selection. With increased packet collisions owing to traffic load, a false link break is assumed at a node that can force the protocol to switch to a cached route or start a new route discovery phase. The semi-proactive nature of LEPR does not allow it to switch to new quality routes that become available as the topology changes, but rather switches to cashed routes [59].

VI. UAVS IN AGRICULTURE

The use of UAVs in agriculture have reduced costs while increasing yields, efficiency and profitability [71] as farmers have access to constant surveillance and data for improved decision-making such as pesticides and fertilizer applications, irrigation, and harvesting. The communication between the UAV-ground nodes is modelled as line-of-sight (LoS) in [72] whereas in real-world scenarios, the vegetation in agricultural fields varies with seasons. The trees shed their leaves in some seasons and they sprout in other seasons, resulting in non-line-of-sight (nLOS) scenarios. The crop, grass and other weed heights also vary with seasons; hence, the grass and weeds will cover sensors buried underground or placed close to the ground surface, creating nLoS or shadowing scenarios. Proper positioning of sensors in the farms was advocated for in [73] as the vegetation can attenuate the radio of each sensor if the sensors are not placed with respect to the seasonal variation of the vegetation. For crop monitoring and management, the UAVs fly close to the plants or crops of interest; requiring algorithms that enable such manoeuvrability. When a swarm of UAVs is deployed in such scenarios, the LoS between UAVs is not guaranteed because the LoS can be obstructed by existing trees between the UAVs as some trees are left or planted on the farms to act as windbreaks. This obstruction destabilizes the routes created between the UAVs by the routing protocol.

A. CHARACTERIZING DATA COLLECTION IN AGRICULTURAL SITES

1) LIVESTOCK MANAGEMENT

Monitoring and managing a large number of livestock in an area are stressful. In [74] and [75], UAVs were used to count and monitor livestock. However, the latter further analyses the limited use of UAV in this area owing to the limited payload, which results in limited flying time. Other drawbacks include landing in rough terrain, and other environmental issues such as high winds, flying birds, and signal loss due to vegetation cover.

2) CROP MONITORING AND LAND USAGE

A drone is used as a remote sensing tool to capture and transmit data for crop monitoring and management [76] as the UAV can capture 'images of high spatial and temporal resolution.' This can help farmers identify areas where crops are infested or weeds grow abnormally. In [12], smart sensors were used to monitor the health of the bees and the yield of honey on the farm from which UAVs were used to collect the data. In [77], a UAV for spraying weed control and pesticides was developed. In the above articles, the application of artificial intelligence is important for identifying infected crops and planing the UAV path. In [78], landscape diversity was classified, and satellite images, together with field surveys, were used to establish the tree species, land use, and land cover. This approach is costly and time-consuming, however, in [79], a UAV was used to classify land use and cover to address low and medium-resolution satellite platforms.

3) SOIL CONDITIONS ACROSS THE FIELD

The soil conditions in terms of moisture and nutrient levels across the farms are often not uniform. Sensors can be embedded in the soil [17] to record the above contents, including soil aeration. This information helps farmers to make decisions regarding which section of the farm requires irrigation and fertilizers. Algorithms can be designed, developed, and implemented to interpret the variation in soil contents and navigate the UAV to the points of interest. However, agricultural fields are usually exposed to extreme temperatures without regulations that affect the sensors deployed, their communications and battery lifetime [80]. Therefore, that requires a routine maintenance of the sensors and their careful placement from extreme temperature locations.

B. UAVS AND ROUTING PROTOCOLS IN AGRICULTURE

1) PATH OPTIMIZATION

Efficient path planning requires efficient communications between UAVs or between UAV and ground nodes. This requires network topology control, good link quality, proper energy optimization, and obstacle avoidance. These aid the routing protocol to work efficiently by establishing

TABLE 4. Metrics and their effects on mobility prediction and the routing protocols.

Parameters influencing routing and predictions	Effect of parameter on protocol and predictions	How to mitigate the effect
Mobility	Frequent changes in topology, link breakages and increase in route discovery processes	1. Velocity, path planning and optimization to reduce total coverage distance [60] 2. Use of prediction methods to estimate node's states such as Position, velocity, direction [61] and traffic load [62]
Energy	Poor route selection and high packet drops [63]	1. Use of reactive routing protocols 2. Route discovery methods that reduce energy consumption [64] 3. Modelling that maximizes the coverage of all nodes (UAV and sensors) rather than fixing cluster heads [65]
Bandwidth	End - to - end delay, packet drops, retransmission and poor QoS [66]	Multipath routing [67]
Link - Quality	Frequent retransmission, packet drops and poor QoS	1. Cross - layer link quality design [68] 2. Prediction methods [69] 3. Multipath routing techniques [70]

the communication path, minimizing control overheads and minimizing end-to-end delays [81].

2) UAV COORDINATION

UAVs collect data in the form of images or text stored in repositories. They also collect images and videos by capturing them as they fly across farms. The UAVs can also be used to spray fertilizers or pesticides. In the mentioned applications, a single UAV or a fleet of UAVs can be deployed. As batteries constrain UAVs, a backup UAV is required for replacement when the UAV in operation runs out of the battery; hence, there is a need for coordination. In [82] when a UAV attains its residual energy, it sends a help message calling for a replacement as it returns to the recharging facility. This help message contains the GPS position at which it halted its operation. This article suggests the help message generated by the exiting UAV includes the IDs of the nodes it has served if it was deployed to serve ground sensor nodes. These approaches will help the replacement UAV to avoid duplication of activities. This coordination requires proper use and design of routing protocols.

VII. CHALLENGES AND FUTURE WORK

The energy used by UAV batteries for communication is usually low compared to that consumed by the motors to drive the UAV forward and overcome certain forces such as aerodynamics. However, for data collection involving ground sensor nodes, the main power consumption by the ground nodes is the energy used to transmit the data compared to that used for sensing. With the metrics considered above, the following are the open issues to be addressed for efficient routing protocols and path optimization in UAV data collection.

A. LINK QUALITY AND NETWORK PARTITIONS

One of the key challenges of FANETs is the successful delivery of packets to their destination. Owing to the dynamic topology, links are unreliable [83], degrading the quality of the signal due to fading, attenuation, and the Doppler Effect [84]. For guaranteed performance, specific metrics such as the signal-to-noise ratio (SNR), expected transmission counts (ETX), link lifetime (LLT) as shown in Figure 6 and residual energy [85] are requirements for modelling to support the routing protocol and mobility prediction. The link quality combines both physical and software metrics; however, the routing protocols focus on improving the network layer without emphasizing the physical and MAC layers that are responsible for collision avoidance and power control of the devices [86]. The use of machine learning to model the MAC and PHY layers metrics can aid the network in adapting to changing conditions thereby enabling efficient operation of routing protocols [87].

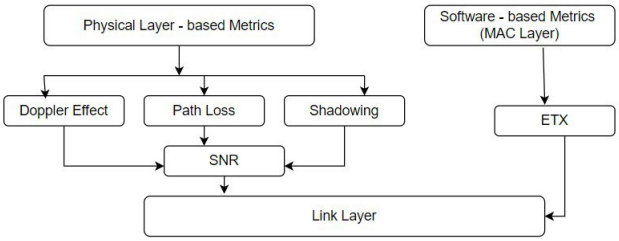

FIGURE 6. Link layer metrics.

In [88], the hierarchical manner of the TCP/IP was designed in NS-3, and the effect of system parameters such as access mechanism, UAV trajectory pattern, UAV velocity, number of sensor nodes, and their traffic generation on throughput and average delay was investigated. However, the study did not relate its findings to the existing routing protocols. In [89], a routing protocol for low-power lossy networks (RPL) was modelled using biographs. The modelling factored in topology changes by varying the radio links between nodes and observed that the RPL could construct a Directed Acyclic Graph (DAG) on a new topology. This was achieved by modelling trickle-timers for repairs in the RPL protocol. This study proposes additional modelling by including the mobility of the nodes. The two sets of results can be compared for consistency. The effects of any variations observed can then be evaluated using a specific protocol.

B. PREDICTIONS

There are design challenges as the dynamic nature of the network results in a reduced network lifetime, self-organization, and high errors. If the network future changes can be precisely predicted, the errors can be minimized significantly as the nodes can send data packets to the destinations when the routing metrics are assured to deliver the packets. The metrics to consider in managing the network's future topology changes are as follows:

1) LOCATION

The Euclidean distance or the root - mean - square error (RMSE) between the current and predicted locations.

$$RMSE = \sqrt{\frac{\sum_i^N (x_i - \bar{x}_i)^2}{N}}, \tag{1}$$

where, x_i = Actual observation time series and $\bar{x}_i$ = Estimated time series. The location information also include the angle of arrival (AoA) of the signal and the UAV's flight direction.

2) SNR

There is a requirement to track the SNR between the current and predicted future location as the lifetime of the link changes with the mobility [90]. Given the direction and acceleration or velocity of flight and the air-to-ground channel modelling, prediction of the horizontal distance d_{Th} at which the link fades to the extent that the signal is not decoded correctly is obtained; therefore, as shown in Figure 7, the reference node knows the lifetime of the link and takes alternative measures when the threshold distance is attained. φ_{UAV} being the heading while θ the elevation angles of the UAV.

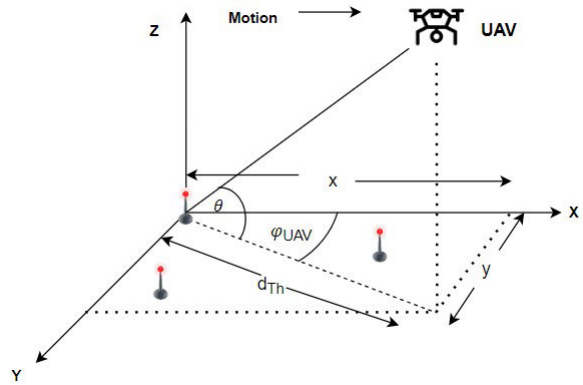

FIGURE 7. 3D location of UAV.

3) RESIDUAL ENERGY

Minimum residual energy is required to allow the sensor node to continue sensing activities or return the UAV to the ground. Providing a meaningful indication of the research direction requires precise prediction of the topology and link state changes using the SNR, location, residual energy, and the available bandwidth as discuss in the next proceedings.

C. POSSIBLE METHODS OF PREDICTIONS

Predicting the network states precisely reduces end-to-end delay, which is an important metric for measuring the effectiveness of routing protocols. There are several prediction techniques found in [91], [92], [93], and [94] and they are categorized here as follows;

- **Deterministic:** For UAV mobility, constant acceleration or constant velocity is assumed. As explained below, it can also be used to predict the remaining useful life (RUL) of a battery.
- **Stochastic** This includes the use of Markov Models and Kalman Filter.

1) DETERMINISTIC PREDICTIONS

This assumes a constant velocity and acceleration, and the prediction is based on the node's observed position, velocity, and acceleration. The current location, speed and direction are used to estimate the future positions. This approach can predict the link expiration time (link lifetime) or remaining life time of a battery. In precision agriculture, sensor nodes are typically exposed to extreme temperatures and no temperature regulations are provided. In [95], a prediction method that incorporates sources of irregularities into an algorithm such that uncertainties are efficiently handled in

a deterministic form was proposed. Noise was used as a source of uncertainty to predict the remaining useful life (RUL) of the lithium batteries. The RUL was found to be approximately 15%. In the aforementioned study, the effect of temperature was neglected; however at temperatures of 0° C the viscosity of the electrolyte increases, which in turn reduces the ionic conductivity. In contrast, high temperatures increase the charge transfer resistance at the electrode-electrolyte interface. The latter causes battery ageing and thermal runaway [96]. The battery's discharge capacity is higher between 25° C and 40° C, with the peak at 40° C [97]. The effects of these temperatures reduce the overall performance of the battery.

2) STOCHASTIC

This prediction method can be used to estimate the future position and/or link quality of a node. For a UAV network with ground sensor nodes, a probabilistic method can be used to track the position and flight direction of the mobile node, including heading angles. The prediction is further divided into Regressive, Autoregressive, Markov, Kalman Filter, and Monte Carlo Methods. In [98] and [99], regressive approaches were used to predict and model the UAV route to a destination for efficient aircraft control. In the first study, the aircraft flight time data obtained from the simulator and were used to perform the prediction. The prediction was achieved using a random forest (RF) model that combined multiple decision trees. A random subset of the available data was used to train each tree. Subsequently, the researchers compared their models with Long Short-Term Memory (LSTM) and logistic regression (LR). The use of real UAV flight data to validate the model is interesting. The latter study uses random tree algorithms for trajectory design to achieve the shortest path and reduce energy. The work also observed the terrain, obstacles, and local-authority-restricted flying zones and altitudes. The work in [100] used UAV flight data and autoregressive approaches to model quality parameters such as final prediction error, residual analysis, and fit percentages as metrics of interest for experimental UAVs. The model provided in this work is consolidated and can be reused by interested researchers to estimate related platforms, such as the location prediction of target nodes. A Kalman filter was used in [101] to jointly estimate the range and direction of arrival (DOA) based on SNR gains. The approach was found to have an outstanding performance for a low SNR while tracking UAVs. This work was limited to a single target. In [102], an uncertainty and error-aware Kalman filter was proposed and used to track UAV missions based on the root-mean-squared error (RMSE) in noisy environments. The RMSE values were calculated based on the SNR values. The article stated that this tracking mode does not guarantee security as different UAVs have different signatures, and hence proposed future works to mitigate signatures from other UAVs to avoid being tracked maliciously. Fanally, [103] used a Continuous Hidden Markov Model (CHMM) with SNR to enhance the spectrum utilization in UAV applications. For practical applications, the imperfect SNR was reconstructed using graphical methods to enhance the CHMM-based spectrum sensing. The model does not fully utilize historical information. There is a need to use a higher-order Markov Chain to modelling spectrum sensing and achieve prediction accuracy. This modelling is suitable for applications that use data centric routing (DCR); however, no routing protocol was used. As the above stochastic predictions were able to precisely predict the required parameters such as SNR, speed, position, range, and DOA, they have not been tested to outline the effects of these parameters on the routing protocols between the nodes in question.

D. HOW THE PREDICTION CAN IMPROVE ROUTING

Based on the analysis above, the current location, speed, environmental disturbances and direction can be used to estimate the position of the UAV at a future time, as well as the SNR of the link/route and link state/link lifetime at a future time. This prediction can help routing protocols to select routes with a better SNR and longer life for guaranteed data packet delivery. A 3D GPSR (Greedy Perimeter Stateless Routing) algorithm using Kalman Filter prediction for UAVs with omnidirectional antennas was proposed in [104]. The Kalman Filter is used to track the location of the UAV, and the algorithm then uses the radiation pattern of the antenna in combination with a flooding strategy to solve the 3D void problem in geographic routing. However, the radiation pattern of the antenna changes with precipitation [105], [106], [107], which this article proposes as one of the open issues for UAV deployment in agricultural sites. The study of [104], [108], [109], and [110] achieved low end-to-end delay, high packet delivery ratio, and low energy consumption.

E. FUTURE WORK

This work proposes the implementation of the prediction methods discussed above for UAV to collect real-world data. Algorithms for path optimization and dynamic adjustments of UAV's speed while tracking transmission requirements regarding sensor nodes' density and placements can be designed, deployed and tested in the real world. The use of multi-user MIMO for the design of UAV network for data collection in areas with high-density nodes can be exploited

VIII. CONCLUSION

Predicting topology change, residual energy, and link lifetime is important in wireless transmissions to avoid unnecessary packet drops. In this study, scenarios for UAV data collection from agricultural sites were reviewed with a focus on establishing an efficient and fault-tolerant network. The main challenges such as limited battery endurance, bandwidth constraints, link intermittency, weather and seasonal variation of the vegetation are analysed including how they impede

UAV's data collection. The effect of mobility, limited energy, link quality and bandwidth on the routing protocols are discussed including antenna selections and their placement on the UAVs. In terms of path planning and optimizations, prediction methods such as stochastic and deterministic are discussed. This study finds that reactive protocols minimize bandwidth usage but takes time to establish a route while proactive protocols execute frequent table updates resulting into high bandwidth and energy usage. However, modification of protocols (such as hybrid protocols), mobility predictions and optimization of UAV's path with regards to crop and ground sensor nodes locations can minimize latency and excessive usage of energy and bandwidth. This, together with proper selection and placement of antenna on UAVs result in a fault-tolerant network and thus effective and efficient data collection. Static routing protocols are generally not suitable for UAV applications but can be used in UAV swarm while optimizing topology control. This is expensive for agricultural applications that require minimization of the number of UAVs deployed.

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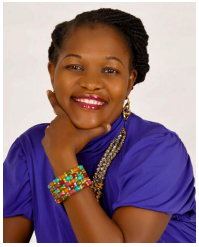

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